

Solution Requirements

Date	30 September 2026
Team ID	6ab3ecd3a238dc999eb3fcc2
Project Name	ComicCraft AI
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	No login required in the current version; users access comic generation directly.	High
2	Authorization levels	No role-based access in the current version; all users use the same creation and export flow.	Medium
3	External interfaces	Connects to Google Gemini APIs and Hugging Face Diffusers/Stable Diffusion for text and image generation.	High

4	Transactions processing	Receives form/JSON input, generates outline → narration/dialogue → panel images → layout → downloadable PDF.	High
5	Reporting	Provides a downloadable PDF comic and JSON layout/PDF path for API clients.	Medium
6	Business rules, etc.	Uses user prompt, character, setting, tone and art style to produce a structured 5-panel comic; users may regenerate with different preferences.	High
7	Compliance to laws or regulations	No specific legal or industry compliance requirement is stated in the project documentation.	Low
8	<i>Other</i>	Shows a comic preview before export and provides an export-success flow after PDF generation.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	Generation speed depends on external AI services; Gemini Flash is selected for fast structured outline generation.	No fixed numeric SLA specified
2	Scalability	Modular FastAPI and AI integration can be extended; current version focuses on single-comic generation.	Future expansion supported; no numeric limit specified
3	Security & Data Privacy	Keep Gemini and Hugging Face credentials in environment variables/.env; do not hard-code secrets.	Secrets stored outside source code
4	Reliability & Availability	FastAPI routes use HTTPException and try/except handling for API/JSON errors.	Graceful error handling for expected failures
5	Usability & Accessibility	Simple HTML/CSS/Jinja2 flow for input, preview and export; designed for non-artists and casual users.	Clear form → preview → PDF flow
6	<i>Other</i>	Modular templates and backend functions support maintenance and future enhancements.	Separate components/routes for core features